

Satrajit Ghosh

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EDUCATION

Rutgers University

Master's in Electrical and Computer Engineering - Machine Learning Track; GPA: 4.0/4.0

- Research in agentic AI within custom C++/OpenGL game engine development
- Experimental analysis of the McGurk effect and its implications on deep learning models

Relevant Coursework: Reinforcement Learning, Multimodal Deep Learning, Parallel and Distributed Computing

Princeton University

Graduate Exchange Program Neural Rendering; Grade: A

- Conducted research on neural rendering pipelines, including NeRF-based avatar synthesis and generative view selection
- Designed pose refinement module using Gaussian Fourier Features and contributed to transformer-based interaction modelling

Institute of Engineering and Management

Bachelor of Technology in Computer Science and Business Systems; GPA: 9.28/10

- President, AR/VR Club; 5th place Intel OneAPI, 1st place IEEE Elevate, 3rd place Google Solving for India
- Co-authored "VR Arena" in Springer Nature; Key coursework: OOP, DBMS, Automata Theory, Digital Electronics

EXPERIENCE

Tech & Product Intern

New York · NY

- Built investment scoring platform for private capital with enriched data pipelines
- Developed AI phishing detection system using behavioral anomaly modeling
- Delivered full-stack prototypes with model integration and competitive research

Grader, Introduction to Machine Learning

New Brunswick · NJ

- Graded homework, quizzes, and exams under Professor Waheed U. Bajwa, developed rubrics and solution keys, and provided feedback and clarification on core machine learning concepts to students.

Graduate Researcher

New Jersey · USA

- Designed and built high-fidelity simulation systems for autonomous vehicles using real-world driving data
- Recreated, and evaluated SafeBench autonomous-driving scenarios on collected real-world datasets.

AR/VR and ML Intern

Kolkata · India

- Built 3D-Model of Ericsson facility using Blender and Unity for Microsoft HoloLens AR
- Created VR House Tour application on Oculus for client procurement and VR annotation tools
- Developed Mandarin Translator app with OpenAI for Indian military using Android Studio

PROJECTS

Neural Character Generation from Unstructured Images

NeRF, Computer Vision, 3D Reconstruction

- Developed pipeline to generate photorealistic, animatable 3D avatars from unstructured image collections
- Refined camera poses using NeRFtrinsics and Gaussian Fourier Features with multimodal tokens (DINOv2 + pose + focal)
- Built transformer-based view selection module and attention-augmented NeRF for high-fidelity reconstruction

LAGEngine Master's Thesis Game Engine

C++, OpenGL, CUDA, Multithreading

- Built a modular 2D/3D engine with real-time physics and parallelized OpenGL rendering across CPU threads and CUDA.
- Integrated a learning-based scheduler that optimizes resource allocation at runtime to reduce frame time and GPU stalls.

VR Exercise Game (Vitality VR)

Unity, Deep Learning, Google Cloud

- Built VR fitness game with web dashboard using Deep Learning deployed on Google Cloud
- Leveraged DNN for user behavior insights and GANs for diet suggestions; 3rd place Google Solving for India

Code and Conquer

Node.JS, WebGL2

- Built gamified coding and productivity platform with real-time WebGL2 tower defense battles and productivity based rewards

Intelligent Traffic Management System

YOLOv5, OCR

- Used YOLOv5 for traffic detection and OCR for license plate recognition; 5th place Intel OneAPI Hackathon

TECHNICAL SKILLS

Languages: C++, Python, C#, Java **Tools:** Unity, Blender, OpenGL, PyTorch, OpenCV

Certifications: Microsoft Azure Fundamentals, Data Science Foundations using R